COUNTABLY INDEXED AFFINE FORMAL ALGEBRAIC SPACES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The delivered context span is the author's own definition of an affine formal algebraic space, which the statement invokes; it is copied whole.

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Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The presentation X = colim X_lambda of an affine formal algebraic space along a directed set. Same chapter, Definition definition-affine-formal-algebraic-space, invoked at statement lines 2429-2431. Fixes the second of the two equivalent conditions and the indexing by a directed set that the proof reindexes.
- Every countable directed set has a cofinal subset isomorphic to (N, >=). Algebra, Lemma algebra-lemma-ML-limit-nonempty (the same observation is used in its proof), invoked at proof lines 2436-2438. Produces the sequential cofinal subsystem that converts one indexing into the other.

Source reference. Original lemma 0597: sources/stacks-chow/tags.txt, source lines 6805–6805. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
   (a) each X_λ is affine,
   (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

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such that

$$X \cong \operatorname{colim}_{\lambda \in \Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let X be an affine formal algebraic space over S. The following are equivalent*

(1) *there exists a system $X_1 \to X_2 \to X_3 \to \ldots$ of thickenings of affine schemes over S such that $X = \operatorname{colim} X_n$,*
(2) *there exists a choice $X = \operatorname{colim} X_\lambda$ as in Definition 0.1 such that Λ is countable.*

Proof. This follows from the observation that a countable directed set has a cofinal subset isomorphic to $(\mathbf{N}, \geq)$. See proof of Algebra, Lemma 0597. □